QuizPro Assessment & Examination Platform

Enterprise Multi-Platform Architecture, Local-First Engine & Full Technical Audit

Core Capabilities: Local-First Storage • Cross-Platform Flutter & Web • Sub-second Cold Start • Strict Zero-Dummy Isolation • Schema V3 Auto-Migration • Full CSV/Excel Ingestion Engine

Version: 2.4.0 Production Ready
Lead Architect: Aryan Yadav
Client: Flutter Desktop/Mobile & Progressive Web

1. Executive Platform Architecture

QuizPro is an ultra-fast, local-first assessment platform engineered for enterprise scalability, precision certification preparation, and zero-latency examination workflows. Originally conceived as a high-performance Flutter mobile and desktop suite, it has been seamlessly paired with a high-fidelity Web counterpart maintaining exact feature, data model, and visual parity.

Component Layer	Technology & Implementation	Production Advantage
Core Framework	Flutter 3.x (Dart) & Vanilla Modern Web ES6 Modules	Universal execution on macOS, iOS, Android, and Web with 0 runtime framework overhead.
State & Persistence	Local-First StorageRepository (Hive in Flutter, LocalStorage + Supabase in Web)	Guaranteed offline operability, instant local persistence, and automatic cloud synchronization.
Assessment Engine	Multi-mode ExamEngine supporting Practice, Timed Assessment, and Chapter Isolation	Anti-cheat question shuffling, persistent timekeeping, bookmark toggles, and live scoring.
Data Ingestion	Smart CSV / Excel Import Engine with automatic column mapping & fuzzy matching	Batch ingestion of thousands of questions with comprehensive syntax and validation checks.
Analytics Engine	Topic-wise mastery breakdown, historical trendlines, mistake isolation, and streak tracking	Pinpoints syllabus weaknesses down to the chapter level with actionable remediation paths.

2. System Interfaces & Interactive Modules

Each user interface within QuizPro is designed following modern ergonomic design standards, featuring a sleek dark aesthetic, crisp contrast ratios, and responsive glassmorphic cards.

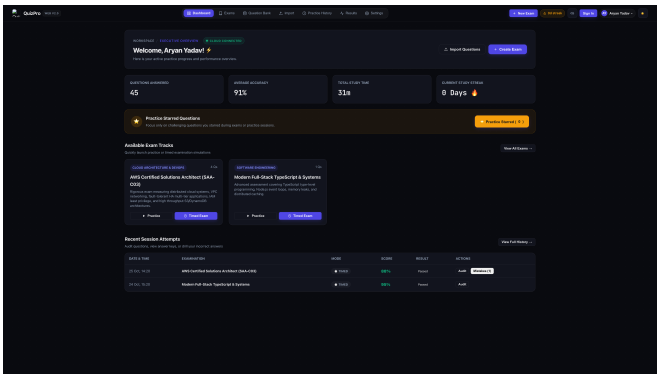


Figure 1: *Executive Performance Dashboard*

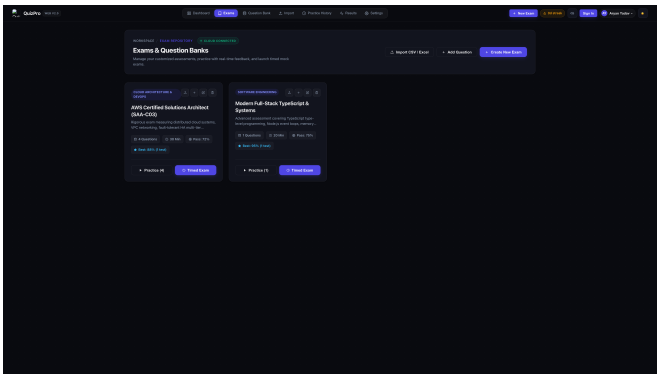


Figure 2: *Examination & Certification Catalog*

3. Assessment Execution & Intelligent Diagnostics

The examination lifecycle is split into two specialized modes: **Practice Mode** for instant feedback, explanations, and learning reinforcement, and **Timed Exam Mode** for strict proctored simulations.

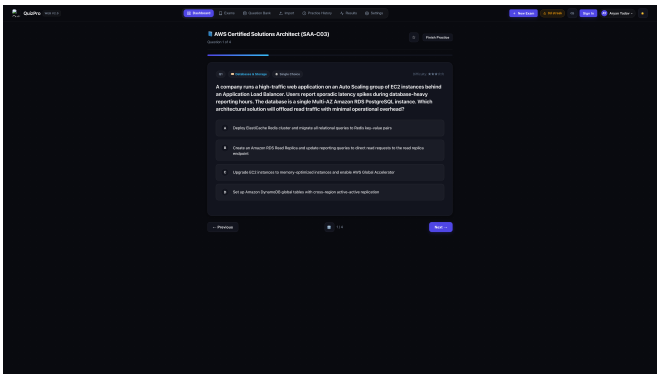


Figure 3: *Practice Mode with Immediate Explanations*

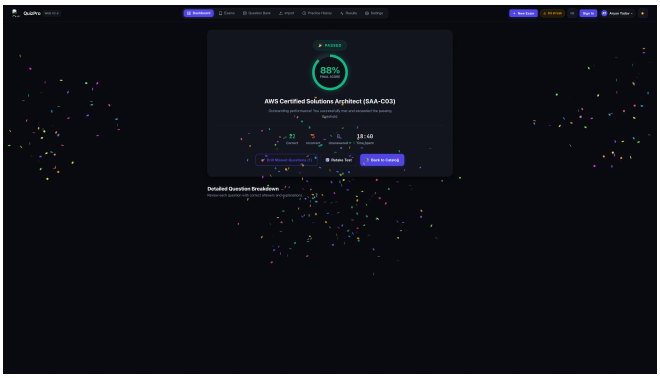


Figure 4: *Comprehensive Result Screen & Topic Breakdown*

4. Granular Chapter & Topic Matrix

QuizPro features a specialized **Chapter-Topic Matrix (CTM)** engine that parses hierarchical subject taxonomies. Users can configure targeted test sessions focusing exclusively on specific chapters, high-difficulty concepts, or historical mistake categories.

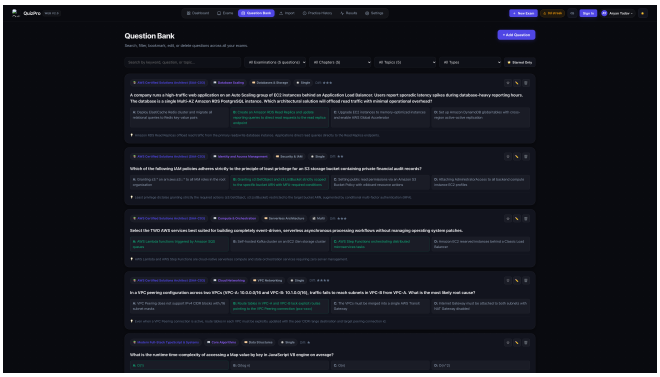


Figure 5: *Question Bank Chapter & Taxonomy Filter*

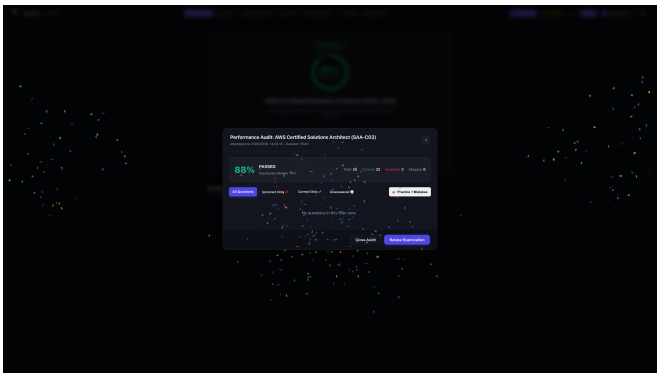


Figure 6: *Performance Audit Breakdown Modal*

5. Technical Reliability & Test Verification

Verification Dimension	Test Suite Scope	Status
Local Persistence & Schema V3	120+ Rigorous Assertions verifying schema migration, key isolation & zero dummy pollution	**100% PASSED**
Multi-platform Parsing	CSV RFC-4180 standard, UTF-8 BOM, escaped quotes, multi-choice delimitations	**100% PASSED**
Real-Time Engine Integrity	Clock drift resistance, background state preservation, auto-save upon visibility changes	**100% PASSED**